The AI Power Gap

How to close the data center-grid timing gap without making communities absorb the risk

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One of the most immediate AI power problems may be a timing problem.

Large data-center projects can seek initial electric service within roughly one to three years. Traditional transmission upgrades can take much longer. That creates an awkward infrastructure race: computing capacity can develop faster than the grid infrastructure needed to serve it.[1]

The scale of demand makes that mismatch harder to ignore. Berkeley Lab's Reference Case estimates that data centers could account for about 11.8% of U.S. electricity consumption by 2030, but that figure sits inside a wide modeled range of 9.5% to 15.3%. The range matters as much as the midpoint: it signals both substantial expected growth and substantial uncertainty about exactly how much power future data centers will need.[2]

The national total also hides the practical problem. An AI developer does not need electricity somewhere in the United States. It needs a very large amount of reliable power at a particular site, on a timeline that keeps the investment viable.

A region may have adequate generation but insufficient transmission. A site may have transmission nearby but need a new substation. Equipment, permitting, interconnection studies, or reliability requirements may become the limiting step.

Having enough electricity is not the same thing as being able to deliver it where and when a data center needs it.

Data centers move fast. The grid does not.

Large data centers are unusually demanding customers.

Connecting one may require utilities and grid operators to study how the new load affects the surrounding system, identify network upgrades, expand substations or transmission, procure specialized equipment, obtain permits, and make sure the connection will not compromise reliability for existing customers.

The data center itself may also arrive in phases rather than all at once. A first phase can seek service on a much shorter timeline than the eventual buildout of a full campus. That gives grid planners some flexibility, but it does not eliminate the underlying mismatch: large-load customers are often asking for substantial initial service faster than conventional network upgrades can be completed.

FERC made that problem explicit in June 2026 when it issued six separate Section 206 show-cause orders, one for each RTO/ISO under its jurisdiction: PJM, MISO, SPP, ISO New England, NYISO, and CAISO. The details differ by region, but the common concern is that existing rules have not kept pace with the size, concentration, and speed of new large loads such as data centers.[3]

DOE is seeing the same pressure from another angle. Its 2026 draft National Transmission Needs Study identifies data centers among the drivers of renewed electricity-demand growth and growing transmission needs.[4]

The result is not one bottleneck but several possible ones: generation, transmission, substations, equipment, permitting, interconnection, and reliability.

The common denominator is time.

Still, closing that gap does not always mean waiting for a brand-new transmission line.

Advanced reconductoring can increase capacity on existing corridors. Grid-enhancing technologies such as dynamic line ratings and power-flow controls can help operators make better use of infrastructure already in place. DOE's 2026 SPARK funding opportunity makes approximately $1.9 billion available for accelerated reconductoring and other advanced transmission upgrades, with an emphasis on projects that can deliver fast, durable grid improvements.[5]

FERC is also actively advancing more flexible approaches to large-load service. Some customers may be able to connect earlier if they are willing and technically able to reduce grid withdrawals under constrained conditions.[6]

That will not work equally well for every AI workload. Some computing can be shifted, throttled, or interrupted more easily than workloads that require highly continuous power. But flexible service expands the options between "wait for every conventional upgrade" and "do not connect."

Developers are also exploring phased connections, new generation associated with their loads, and behind-the-meter generation or storage. A recent Pennsylvania project, for example, paired a planned data-center campus with its own gas-fired generation rather than relying only on a future grid connection.[7]

None of these tools eliminates the need for conventional grid investment. They do make the problem more interesting than simply "build more transmission."

The queue has a credibility problem, too.

Utilities do not just need to know how much power a project wants. They need to know whether the project is real.

A developer considering several possible locations can have legitimate reasons to ask multiple utilities about power availability before choosing a site. But speculative requests can create serious planning problems when very large loads enter study processes without strong evidence that they will ever materialize.

Forecasts can become distorted - and scarce interconnection-study resources wasted - if effectively the same project appears in multiple places or if developers reserve planning attention before making meaningful commitments.

FERC has identified this as a central problem in its 2026 large-load actions. Commissioner David Rosner described speculative requests as clogging queues, diverting resources, distorting forecasts, and forcing grid operators to spend time studying projects that may never be built.[8]

That is why the current reform effort is not simply about protecting residential customers from costs. It is also about making the planning process itself work better.

FERC is pushing regional grid operators toward requirements that become more demanding as large-load projects advance. Rather than allowing a project to consume the same amount of planning attention from beginning to end, regulators can require progressively stronger evidence that it is moving toward actual construction.[8]

That evidence does not have to be reduced to one arbitrary national threshold. Depending on the stage of development, it could include site control, financial security, long-term service commitments, major equipment or supply commitments, or other objective evidence that a project is becoming harder to walk away from.

A useful principle is:

The closer a project gets to consuming scarce grid capacity and planning resources, the more it should have to demonstrate that it is real.

Someone has to own the downside.

This matters because demand forecasts are uncertain while grid infrastructure is not.

Berkeley Lab's 2030 scenarios span almost six percentage points of total U.S. electricity consumption. That does not make the Reference Case useless. It means decision-makers should treat it as a modeled scenario rather than a settled fact.[2]

Individual projects carry even more uncertainty. They can be delayed, downsized, relocated, phased differently, or canceled altogether.

Transmission lines, substations, and other major grid investments are much less flexible. They are expensive, long-lived, and tied to specific places.

If a utility builds substantial infrastructure because a large private customer says it is coming, and that demand later disappears, the infrastructure bill does not disappear with it.

Who should bear the risk that a private developer's forecast turns out to be wrong?

My starting point is that developers should bear most costs and risks reasonably attributable to the demand their projects create.

That phrase is intentionally less tidy than "project-specific." In practice, cost allocation is difficult. A transmission upgrade triggered by one enormous customer may also improve regional reliability, expand capacity for future users, or create broader economic value. Utilities, developers, regulators, and customers can reasonably disagree over how much of an upgrade belongs to one project versus the wider system.

That boundary has to be adjudicated rather than assumed. But ambiguity is not a good reason to default to socializing the risk.

Developers asking utilities to make major investments around their projects should have meaningful exposure if those projects fail to materialize. FERC's June action moves in this direction by requiring regional grid operators to address Cost Recovery Agreements designed so large-load customers pay their applicable share even if they do not come online as planned, reducing the risk that stranded costs shift to existing residential customers.[9]

The principle does not need to be that developers pay for everything. It should be that private projects retain a substantial share of the private risk they create.

There is no single regulator with a "go faster" button.

The policy bargain operates across more than one level of government.

FERC can shape rules involving interstate transmission service, regional planning, study procedures, cost recovery, and large-load access within its jurisdiction.

States and local governments control many other decisions that can determine whether a project actually moves forward: retail electricity structures, generation and resource decisions, local siting, land use, environmental review, and permitting.

FERC Commissioner Lindsay See emphasized that the Commission's role should complement, not displace, state authority.[10]

That matters because a faster federal transmission process cannot, by itself, create a unified national fast track for data centers. A developer might clear one bottleneck and encounter another.

The realistic goal is coordination: clearer and faster federal processes where federal jurisdiction applies, paired with state and local rules that identify requirements early enough for developers to plan around them.

Speed should be earned.

Credible projects should get faster paths to power through better queue management, flexible service where technically appropriate, faster use of existing grid capacity, and more predictable study and approval processes. In return, developers should demonstrate enough project and financial readiness that speed does not become a subsidy for speculative demand.

That bargain can improve both speed and risk allocation: weak or duplicate projects consume fewer planning resources, while projects that are actually ready to build become easier to identify and prioritize.

Community opposition is an emerging implementation risk.

Financial risk is not the only reason developers should be expected to take local impacts seriously.

Large data centers can generate concerns about electricity costs, water use, noise, land use, emissions, and the scale of infrastructure required to support them. Those concerns do not automatically make a project undesirable, nor are they evidence that community opposition is the dominant national constraint on AI infrastructure.

But recent events show that local resistance can become operationally important.

Indianapolis approved a moratorium on new data-center development on August 19, 2026, following local opposition.[11]

The previous day, Pennsylvania removed data centers from its Fast Track permit program and added stronger environmental, transparency, and local-approval requirements.[12]

Those events are too recent to prove a settled national trend. They are better understood as signals of an emerging implementation risk.

A technically viable project can still lose time if the surrounding community believes it is being asked to absorb disproportionate costs or resource impacts. That makes early disclosure and mitigation useful for more than public relations.

Developers should identify major local impacts early, explain them clearly, and bear reasonable costs associated with impacts directly attributable to their projects. Communities should have a meaningful opportunity to raise concerns before enormous infrastructure commitments have already been made.

Ignoring predictable opposition can be a slow way to build quickly.

Build faster. Make the bet better.

The U.S. power challenge created by frontier AI is real. But one of its most immediate forms is a timing mismatch between fast-moving large loads and slower-moving electric infrastructure.

The response should match the problem: use existing grid capacity more intelligently where possible; improve queue and readiness rules so scarce planning resources go to credible projects; coordinate federal, state, and local decisions; and keep developers meaningfully exposed to the costs created by their own forecasts.

The United States has legitimate economic and national-security reasons to preserve the ability to scale AI infrastructure quickly. It does not follow that existing customers and surrounding communities should place an open-ended financial bet on every private forecast of future compute demand.

That will not make the grid move at software speed.

But it may help decision-makers answer a more useful question: Which projects are worth moving faster for?

Notes

1. Federal Energy Regulatory Commission, PJM Interconnection, L.L.C., Docket No. EL26-67-000, Order Instituting Proceeding Under Section 206 of the Federal Power Act (June 18, 2026). Source

2. Lawrence Berkeley National Laboratory, United States Data Center Energy Usage Report: 2025 Update. Source

3. Federal Energy Regulatory Commission, FERC Launches Aggressive Targeted Action to Speed Large Load Integration (June 18, 2026). Source

4. U.S. Department of Energy, Office of Electricity, DOE's Office of Electricity Publishes 2026 Draft National Transmission Needs Study to Strengthen America's Grid (July 9, 2026). Source

5. U.S. Department of Energy, Office of Electricity, Speed to Power through Accelerated Reconductoring and other Key Advanced Transmission Technology Upgrades (SPARK) (March 12, 2026). Source

6. Commissioner David Rosner, Federal Energy Regulatory Commission, Remarks on the Large Load Show Cause Orders, E-7 to E-12 (June 18, 2026), discussing flexible/non-firm transmission service and alternatives for serving large loads. Source

7. Reuters, Alpha Compute to buy Pennsylvania land, gas rights for $55 million data center campus, company says (Aug. 11, 2026). Source

8. Commissioner David Rosner, Federal Energy Regulatory Commission, Remarks on the Large Load Show Cause Orders, E-7 to E-12 (June 18, 2026), discussing speculative requests and project-readiness requirements. Source

9. Commissioner David Rosner, Federal Energy Regulatory Commission, Remarks on the Large Load Show Cause Orders, E-7 to E-12 (June 18, 2026), discussing Cost Recovery Agreements and stranded-cost protection. Source

10. Commissioner Lindsay S. See, Federal Energy Regulatory Commission, Remarks on the Large Load Show Cause Orders, E-7 to E-12 (June 18, 2026), discussing complementary federal and state authority. Source

11. Axios Indianapolis, Indy data center moratorium unanimously approved (Aug. 19, 2026). Source

12. Reuters, Pennsylvania governor signs order imposing new rules to set up AI data centers in state (Aug. 18, 2026). Source